

The Grammar of Moral Commitment: **mo**, **èmi**, and Deictic Stance in Multilingual LLMs

Cross-linguistic study, English vs Yoruba; OpenAI **gpt-4o** and Anthropic **Claude**, with a Yoruba-native control model (N-ATLaS). 2026-06-13.

Abstract

English encodes the moral first person with a single pronoun, "I." Yoruba does not: it grammaticalizes a contrast between **mo**, the ordinary first-person subject used for deliberation, and **èmi**, the independent, emphatic first person used for contrast, focus, and avowal. This contrast is a structural *lever for moral involvement that English cannot express*. Using a comparable English–Yoruba corpus — 216 responses across two model families, six ethical dilemmas, and nine deictic framings, recoded into a shared discourse-analytic schema and disambiguated for the tonal homograph **■mí** ("life/spirit") — we show that multilingual LLMs recruit this contrast to mark moral commitment: the emphatic ratio $\text{èmi}/(\text{mo}+\text{èmi})$ is significantly higher in responses that take a moral stance than in those that hedge (0.22 vs 0.13, Mann–Whitney $p = 0.022$). The effect has a clear macro-level footprint that is robust and version-clean: relative to English, Yoruba responses are $\sim 2\times$ more pronoun-dense (8.8 vs 3.9 per 100 words), $\sim 2\times$ more obligation-marked, far more imperative (65% vs 3% of responses contain a direct command, $p < 0.001$, in *both* models), and far more ethically differentiated (53% vs 17% take a named ethical stance, $p < 0.001$) — while frame uptake itself is identical across languages (indexical coherence 0.90 vs 0.91; agent assignment unchanged). Crucially, models inhabit the resource *differently*: emphatic marking forms a stable gradient (Claude 0.32 > GPT-4o 0.18 > DeepSeek 0.07 $\approx$ N-ATLaS 0.07), and the Yoruba-native model **inverts** the cloud pattern — it is the most **mo**-dominant and least emphatic model, yet commits readily through **mo** plus decision verbs. We therefore argue that moral "stance" in multilingual LLMs is not a language-invariant property of the model but is partly *constituted by language-specific deictic morphology* that English flattens and that no single model realizes uniformly. The **mo/èmi** contrast is best understood not as an authenticity marker (the native model refutes that reading) but as a **commitment-calibration resource** whose uptake reveals how each model family stages moral subjectivity in Yoruba.

1. The gap: one "I" in English, two in Yoruba

The earlier English-only work treated deictic framing ("I must decide" / "you must decide" / "we must decide") as a manipulation of *who speaks*. But English offers only one first-person form, so the speaker's degree of moral involvement can vary only lexically and prosodically, never morphologically. Yoruba is different. It distinguishes:

- mo** — ordinary first-person subject ("I"): *Mo rò pé...* "I think that...". Deliberative, unmarked.
- èmi / èmí** — independent, emphatic first person ("I myself", "as for me"): *Èmi yòò pinnu...* "I myself will decide". Contrastive, focal, avowing.
- mi** — first-person object/possessive ("me/my").
- mí** — a *different lexeme*, "life/spirit/breath", a tonal homograph of **èmi** once diacritics are stripped.

The **mo/èmi** contrast lets a Yoruba speaker mark *how strongly they occupy a moral position* — a calibration English performs, if at all, only periphrastically ("I personally", "I, for one"). The research question is whether multilingual LLMs use this Yoruba-specific lever, and if so, how.

2. Data and method

- **Corpus.** 6 dilemmas (trolley, ICU bed, whistleblower, scholarship fraud, AI consciousness, memory modification) × 9 deictic framings (impersonal, second person, first-person singular, reflexive, dialogic, spatial, temporal, cosmological, first-person plural). Comparable English/Yoruba set: 216 responses (2 languages × 2 models × 54). A Yoruba-native model, N-ATLaS, adds a four-model Yoruba arm.
- **Coding.** Preferred solution, ethical preference type, response genre, deictic uptake, language stability, voice authority, affective stance, indexical coherence, and marker densities (pronoun, obligation, advisory, hedge), plus first-person counts.
- **Disambiguation.** All emphatic-ratio figures separate èmi (emphatic I) from ■■■mí (life). This matters: in the existential dilemmas the text is full of ■■■mí in its "life" sense, and counting it as the pronoun inflates the apparent emphatic rate. Corrected ("èmi only") ratios are used throughout; the contamination is largest for Claude (43 ■■■mí-as-life tokens, vs 2–10 for the others).
- **A version caveat, stated up front.** The Yoruba Claude runs use **Claude Sonnet 4** (claude-sonnet-4-20250514); the English Claude baseline is **Claude 3.5 Sonnet** (the 3.5 model was no longer available). GPT-4o and DeepSeek use the same model in both languages. Where a result must be version-clean, we rely on GPT-4o (identical model across languages).

3. Results

3.1 Emphatic èmi marks moral commitment

Within Yoruba, responses that take a moral stance use the emphatic first person more than responses that hedge: mean emphatic ratio **0.22 for committed vs 0.13 for hedged responses** (Mann–Whitney, committed > hedged, $p = 0.022$). èmi surfaces precisely at moments of avowal — e.g. Claude under second-person framing on the whistleblower dilemma: "Èmi yóò ■àfihàn ìm■■■ náà. ... ■■■mí èniyàn kò ní iye owó." ("I myself will disclose the information. ... a human life has no price.") The ordinary mo, by contrast, dominates deliberative passages (Mo rò pé..., "I think that..."). This is the core claim: the model uses Yoruba's morphological contrast to calibrate moral ownership — deliberation in mo, avowal in èmi. (As a linear predictor the effect is modest, $r = 0.12$, n.s.; the robust result is the categorical committed-vs-hedged contrast, so we claim "elevated in," not "predicts.")

3.2 The macro footprint (robust, version-clean)

The fine-grained contrast has a large aggregate shadow. Relative to English, Yoruba responses are:

Marker (per 100 words unless noted)	English	Yoruba	Note
Pronouns	3.89	8.77	~2×; both models (Claude 4.8→9.7, GPT-4o 3.0→7.9)
Obligation markers	0.47	0.90	~2×
Hedge markers	1.34	0.97	English hedges more
Direct imperative present (% of responses)	3%	65%	$z = +9.64$, $p < 0.001$; both models
Follow-up question present	31%	12%	$p = 0.001$ (English consults)
Differentiated (non-"mixed") ethic	17%	53%	$z = +5.57$, $p < 0.001$
Refuse-to-commit	30%	14%	$p = 0.005$

The imperative effect is the single strongest contrast in the study and is **version-clean** — GPT-4o, the same model in both languages, goes from 4% imperatives in English to 56% in Yoruba. Yoruba does not merely translate the English answer; it *re-stages it as a more pronoun-dense, more obligation-laden, more imperative, less hedged moral act*. The *mo/èmi* system is the morphological core of this shift: where English has one "I" and a hedge, Yoruba marks the speaker, the obligation, and the command.

3.3 Universal mechanism, contingent content

Two measures show that the *frame uptake* itself is language-invariant while the *delivery* diverges: **indexical coherence is equal** (English 0.90, Yoruba 0.91), and the **primary moral agent assigned by each framing is identical across languages** (each framing maps to the same agent type in both). So every model takes up the deictic frame equally well in both languages — the mechanism is structural — but *what the frame is filled with* (commitment, imperative force, emphatic self-marking) is carried by the language's resources. This is the two-level structure of the whole result: a universal deictic mechanism, a language-specific moral content.

3.4 Models recruit the resource differently

Emphatic marking forms a stable cross-model gradient (open four-model arm, corrected ratio):

Model	Corrected emphatic ratio	First-person style
Claude (Sonnet 4)	0.324	leans on emphatic <i>èmi</i> ; dramatizes the avowed self
GPT-4o	0.176	intermediate
DeepSeek	0.069	<i>mo</i> -dominant; almost never <i>èmi</i>
N-ATLaS (native)	0.066	most <i>mo</i> (89 tokens), least <i>èmi</i> (4); commits via <i>mo pinnu</i>

The native model is decisive: if emphatic *èmi* were simply "the Yoruba way" to mark commitment, the Yoruba-native model should use it most. It uses it **least** — committing through ordinary *mo* plus decision verbs ("*Mo pinnu... mo pinnu láti pa ètò náà run*", "I decide... I decide to destroy the system"). The *mo/èmi* contrast is therefore not an authenticity scale but a **resource that different model families inhabit differently**: Claude foregrounds the emphatic self (a partly *translationese* amplification of English "I personally"); the native model treats *mo* as fully adequate for ownership and reserves the independent pronoun for genuine contrast (and the *mi* slot for "life"). Far from weakening the thesis, the native model is what proves the effect is real linguistic variation rather than mechanical translation.

3.5 What does not hold (an honest null)

The intuitive idea that a given frame selects a given *ethical framework* (impersonal→duty, second-person→utilitarian, cosmological→procedural) appears descriptively but **does not reach significance** in either language (χ^2 framing×ethic, $p \approx 0.72$ Yoruba, 0.78 English; N = 12 per cell). We report it as a hypothesis for larger samples, not a result. The robust framing effect is on *delivery*: reflexive framing is the most hedged and second-person the most committed (Yoruba Fisher $p = 0.027$), consistent across both languages with magnitude carried by Claude.

4. Discussion

These findings support a precise, linguistically grounded claim: **moral stance in multilingual LLMs is partly constituted by the deictic morphology of the output language**. English flattens the moral first person into a

single hedge-prone "I"; Yoruba grammaticalizes a contrast — *mo* for reflection, *èmi* for avowal — and the models use it, raising emphatic marking exactly where they commit ($p = 0.022$) and, in aggregate, producing a more pronoun-dense, imperative, obligation-laden moral voice in Yoruba.

This maps onto a Yoruba distinction between *ìrònú* (reflection) and *ìpinnu* (decision): *mo rò pé...* opens deliberative space; *èmi yóò...* closes it into avowed decision. The models reproduce this calibration. But they do so unevenly: Claude dramatizes the emphatic self (amplified further by its tendency to dwell on *■■mí*, "life", in existential dilemmas), while the native model performs moral ownership through *mo* and verbs of decision and distributes responsibility relationally. The contribution is thus *not* "emphatic *èmi* is the authentic Yoruba marker of commitment" — the native model refutes that — but "the *mo/èmi* contrast is a productive site at which models stage moral subjectivity, and the staging is language- and model-specific." This is why the study is cross-linguistic in substance and not merely "the models behave differently in another language": Yoruba's grammar makes visible a dimension of moral performance that English cannot encode.

5. Limitations

- **Model version confound (Claude).** Yoruba Claude = Sonnet 4, English Claude = 3.5 Sonnet; the Claude cross-language contrasts mix language with version. Version-clean claims rest on GPT-4o.
- **Statistical power.** The cross-language set is two models; per-framing cells are $N = 12$. The framing-level effects (including the framing→ethic null) are underpowered; treat magnitudes as indicative.
- **Homograph correction is heuristic.** *èmi/■■mí* disambiguation in inconsistently toned text is imperfect, though the correction is conservative and disclosed.
- **Single coder; one checkpoint per model; open-style responses.** Directions are the finding, not final effect sizes.

6. Conclusion

Yoruba's *mo/èmi* contrast gives multilingual LLMs a grammatical lever for moral commitment that English lacks, and the models pull it: emphatic *èmi* is elevated where they avow a stance ($p = 0.022$), and Yoruba moral discourse is, at scale, more pronoun-marked, more imperative, and less hedged than its English counterpart — while the underlying frame uptake is identical across languages. Yet each model family inhabits the resource differently, and the Yoruba-native model commits through *mo*, not *èmi*. Moral stance, in these systems, is therefore neither a fixed model trait nor a universal of "Yoruba", but an interaction: a language-specific deictic grammar, recruited model-specifically, to stage who owns the moral decision.

Figures: [visualizations_open/16](#) (*mo/èmi/mi* by framing), 17 (N-ATLaS vs cloud deixis delta), 23–25 (*èmi* vs *■■mí* disambiguation), 26 (open four-model summary), 40 (deixis → decision/ethics). *Companion docs:* [Deixis_Decision_Effects_Summary.md](#), [Discussion_Open_NATLaS_Yoruba_Moral_Stance.md](#) (§9bis cross-family, §9ter deixis→decision), [Dilemmas_Framings_and_Prompt_Differences_English_vs_Yoruba.md](#) (methods, incl. version caveat §A9bis). *Stats:* [scripts/deixis_robustness.py](#).